

Praveen Voonna

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Backend engineer with 6+ years building production-grade distributed systems in Go and Java — low-latency, high-availability services running at scale. Comfortable owning features end-to-end from design through implementation, benchmarking, test automation, deployment and production debugging. Strong on goroutine/channel concurrency patterns, REST and gRPC API design, and both relational (Oracle, MySQL) and document (MongoDB, DynamoDB) data stores. Currently serving notice period — available to start immediately.

EXPERIENCE

Software Engineer · Cisco Systems · Pune

January 2024 – Present

Backend engineer on Cisco's 5G Mobility infrastructure team — the shared control-plane platform underpinning Cisco's mobile-network services deployed at tier-1 telco operators. Own multiple features end-to-end across a Go-based microservices monorepo running on Kubernetes.

- **Designed and shipped a control-plane decoupling change** isolating the management/RPC plane from data-plane interface state, ensuring CLI and operational tooling stay available during network-interface failures — cross-cutting refactor across multiple Go services and tests.
- **Led the IPv6-only deployment-topology feature** — redesigned service bring-up to handle interfaces without any IPv4 address, including configuration schema, startup sequencing and end-to-end tests.
- **Designed a retry/fallback framework for a distributed routing subsystem** with exponential backoff and a static-route safety net, eliminating a class of intermittent install failures observed in production.
- **Owned hardening of a real-time fault-detection subsystem (BFD)**: ephemeral-port allocation per RFC 5881 with randomized→sequential fallback on collision, bounded-retry semantics, and a memory-cache layer that reduced control-plane watch traffic to the etcd cluster.
- **Implemented end-to-end QoS classification (DSCP) for SCTP transport** across config, socket layer, server/client paths and tests — ~2.3k LOC across 26 files; used to differentiate NF traffic classes in production.
- **Reliability work**: switched a hot-path policy update from full reset to incremental reconciliation, fixed concurrency defects in a shared map under high-fanout updates, and removed a startup nil-panic in the metrics pipeline.
- **Platform owner for stack upgrades** across 8+ repositories — Go 1.22, Kubernetes 1.32.1, gRPC v1.79.3, base OS image, datastore client — coordinating with multiple downstream teams.
- **Production ownership**: primary-rotation on-call for the subsystem; resolved 25+ customer escalations spanning routing, fault-detection, transport and metrics, with patches backported across multiple release branches.
- **Quality**: lifted unit-test coverage of a ~150k-LOC Go infrastructure codebase from 93% to 99% by targeting error paths, concurrency edges and CLI handlers — the hard last-mile coverage that catches real regressions before they ship.

Senior Software Engineer · LTIMindtree · Hyderabad

February 2022 – January 2024

Backend engineer on a real-estate platform (Anywhere/Realogy) used by thousands of agents. Owned multiple Go services on AWS, with end-to-end responsibility for design, performance, security and on-call.

- **Benchmarked and cut search API p95 latency from ~1.5 minutes to ~20 seconds** by redesigning the query path, removing N+1 calls and parallelising downstream fan-out using goroutine and channel patterns.

- **Reduced a critical Oracle → MongoDB ETL pipeline runtime from 7 days to 6 hours** via a goroutine-based worker-pool refactor and batched I/O — a ~28× improvement that unblocked daily reporting.
- **Designed and maintained 100+ REST and gRPC endpoints across 10–15 service repositories;** integrated relational (Oracle) and document (MongoDB Atlas) data stores behind a uniform service layer; introduced field-level access control and hardened authn/authz flows across the platform.
- Wrote unit, integration and end-to-end tests; participated in code reviews and on-call rotations across time zones.

Systems Engineer · Infosys Limited · Hyderabad

October 2019 – December 2021

- Built backend and frontend features for major US utilities (PG&E, APS) using Java Spring Boot, ReactJS and Sitecore; shipped customer-facing dashboards and CMS-integrated experiences.
- Top-rated trainee (91%) in the Infosys full-stack engineering programme.

SKILLS

Languages: Go · Java · Python · JavaScript · SQL

Backend & Distributed Systems: gRPC · REST · Microservices · Concurrency (goroutines, channels) · Distributed state (etcd) · Event-driven systems · Observability / metrics

Cloud & Platform: Kubernetes · Docker · Helm · AWS (ECS, EC2, S3, Lambda, Kinesis, CloudWatch) · Linux systems programming

Databases: MongoDB Atlas · Oracle · MySQL · DynamoDB

Networking (specialty): BGP · BFD · SCTP · TCP/UDP · IPv4 / IPv6

AI-assisted development: Claude (Anthropic) · Augment Code — daily driver for test authoring, refactoring and code review on a large production Go codebase.

Practices: System design · Benchmarking · Code review · On-call / production debugging · CI/CD · Agile

PUBLICATIONS

Purpose Fine-Tuned Large Language Model for End User Application Detection

Technical Disclosure Commons, May 2026 · Co-inventor with R. Subramanian, D. Sharma, K. P. Gowrish · tdcommons.org/dpubs_series/9999

AppLLM — a system combining Chain-of-Thought training with LoRA-based fine-tuning for explainable network application classification, integrated directly into the Vector Packet Processing (VPP) data plane to address limitations of traditional Deep Packet Inspection on encrypted traffic.

EDUCATION

B.Tech, Computer Science Engineering (Honours) — Indira Gandhi Institute of Technology (IGIT), Sarang · Biju Patnaik University of Technology, Odisha · 2015–2019 · CGPA 8.34